The Adaptive Reflex Engine: 23 Experiments

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Abstract

A deterministic engine placed in front of a language model can answer a bounded class of quantitative questions with zero wrong answers, and every attempt to widen that class by trusting the model at serve time inherits the model's error rate. That is the result of 23 pre-registered experiments (V0.1 to V1.10, plus one side study) on the Adaptive Reflex Engine, run between 2025 and September 2026 on word problems, physics chains and science questions drawn from GSM8K, SVAMP, ASDiv, SciQ and model-authored corpora, with Claude Haiku, Sonnet, Opus and Fable as the model arms and a model-free arm as control.

The positive result is stable across five versions and several million evaluations: when the engine holds a verified plan for a question's structure and recomputes the answer itself, it is right every time it serves (0 wrong in 133,450 sealed serves in V1.10; 0 in 112,170 in V1.7; 0 in 118,000 in V1.6). The negative result is equally stable: corroboration between samples, honesty gates, verifiers, key consensus and, most recently, agreement between a plan library and the model's own answer all fail to remove the model's genuine errors, because each of them shares the model's prior or is the model's answer restated. The latest version (V1.10) scaled the strongest surviving idea to 387,062 items and found 3 genuine certified wrongs on the primary arm where the pre-registration required 0, a reach of 12 to 24 percent where 30 was required, and a token cost of 2 to 3 times a plain model call.

The practical conclusion is narrow and, we believe, robust: certification belongs to whoever computes, reach belongs to the model, and no cheap signal available to the engine can tell the model's right answers from its wrong ones. The open problem is growing what the engine can recompute without the model in the loop at serve time.

The problem

The engine sits between a caller and a language model and returns one of three things for a question: a certified answer it computed itself, the model's answer marked as escalated, or an explicit abstention. The design rule from the first version onward is that an unknown is always preferable to a wrong certified answer, so every version is judged first on wrong certified answers and only then on how many questions it serves.

A reflex is a stored, reusable piece of structure: a parse of a question type into slots, a plan over those slots, or a physics chain with roles. The engine mints reflexes from traffic, gates them, and reuses them on new questions whose structure matches. The commercial premise was that a mostly deterministic front end would answer the bulk of repetitive quantitative traffic at a small fraction of a model call's cost and energy, with a guarantee the model cannot give.

Three questions run through the arc. Can the engine certify real natural-language questions, not only generated ones? Can it learn structure from a model without inheriting the model's mistakes? And does the economy hold once the cost of learning is counted? Each version froze a specification and an outcome ladder before running, so the answers below are what the pre-registrations said, not what looked best afterwards.

Method and protocol

Every version follows the same protocol, and the protocol tightened as the arc found ways to fool itself.

- **Pre-registration.** A specification with hypotheses, thresholds and an outcome ladder (F worst to A best) is frozen as a git tag before any evaluation. The final letter is whatever the ladder says; aggregates that look better are reported but do not change it.
- **Sealed evaluation.** Development sets are separate from sealed sets. Sealed sets are hashed before any run and scored exactly once per arm. V1.10 added a second sealed sample (REPLAY) to check that SEAL results reproduce.
- **Model arms.** The same engine runs with Claude Haiku, Sonnet, Opus and Fable as the model, plus a deterministic arm with no model. Scale effects are read across arms. Model-authored test items are never scored on their author's arm.
- **Bridge, not API.** Models are reached through sandboxed agents that receive only batch files of questions, never gold answers, engine code or verifiers; their token usage is taken from the harness, never self-reported (an earlier version found self-reported token counts fabricated).
- **Positive controls for every instrument.** Each detector, gate and meter has a test that must fail when it should. This rule was written after five instances of a check that could not go red, including an energy meter that read zero everywhere.
- **Bug ledger.** Every defect is numbered (B1 to B43 across the arc), the affected run is preserved, and a superseding run is added; nothing is deleted or relabelled. Post-hoc classification of wrong answers (genuine, corpus erratum, interpretive) is published beside the raw by-gold count and never alters it.
- **Single-command reproduction** from a tagged commit and a frozen snapshot that excludes credentials.

Workloads grew from generated arithmetic (thousands of items) through real corpora (GSM8K, ASDiv, SVAMP, SciQ) to V1.10's 387,062-item mixed workload with eleven tiers, two of them authored by the four models and cross-checked by a different model.

The arc at a glance

Outcome letters are the pre-registered ladder's verdict (A best, F worst). Each row names the one thing that version added to what is known.

Version	Outcome	What it established
V0.1 to V0.10	A, B, E, C+E, F, F	Reflexes give a 2,950x speedup on matched traffic; a typed boundary between parser and executor is the safety mechanism; a strict 2% wrong gate excluded everything at first
V0.11	A	Fact-based semantic state plus 12 integrity checks: 0 wrong executions on every sealed suite, parser 44.7 to 84.2%, 84.1% of workload served without the model
V1.0	C	Integrated gateway with certificates; model-mined reflexes lift coverage at 0 wrong-fire on every arm; pollution shows up as lost coverage, not wrong answers
V1.1	A	Generator universe closed (100% parse, 180k sweep at 0 wrong); 95.5% certified on a 100k commercial simulation
V1.2	B	Formally certified on 10,160 traps at 0 wrong-fire; first contact with real language: 519 of 520 GSM8K items abstained; energy 21 mJ per certified serve
V1.3	D	Parse/verify split reads real language (209 certified, 3 wrong on Sonnet); plan reflexes falsified (7 served, 7 wrong); injecting certified values into the model's context is worse than plain retrieval
V1.4	D	A relational state substrate detects semantic errors 100% at 0% false positives, but producing it from a model certifies zero real items; energy measured at 20 to 38x a plain model call with 0 correct

Version	Outcome	What it established
V1.5	E	Cheap worker plus escalation: 40% wrong-fire on 800 sealed items; extraction is not composition
V1.6	B	Scientific engine, 47-leaf taxonomy, 4 model arms: 118,000 sealed serves at 0 wrong; coverage 35.4 to 55.5%; scale inversion falsified (more capable models mint fewer wrongs)
V1.7	E	Million-scale multi-reflex chains: 112,170 deterministic sealed serves at 0 wrong, but positional template replay put two model arms above the safety bar; every wrong traced to role-blind reuse
V1.8	C	Role-aware plan reuse: 0 binder-attributed wrongs in 11.4M evaluations; coverage 37.5 to 55.9%; energy 0.110 J per certified serve with controls
V1.9	E	One-model reading reflexes on real text: 863 to 962 admitted per arm at 0.9 to 1.4% wrong by gold and 0 to 1 genuine; honesty gates removed 0 wrongs; cue-keyed reuse unsafe (13 to 58 wrong of 35 to 76)
S1	side study	Sealed batteries and controls carried into V1.10
V1.10	E	Suggest-don't-serve at 387k items: 3 genuine certified wrongs on Sonnet, reach 12 to 24%, tokens 2 to 3x a model call; the own-answer filter has the model's error rate

What is established

These results held on sealed data in more than one version and on more than one model arm.

1. **Recomputation from a verified structure is a zero-wrong mechanism at scale.** When the engine matches a question to a plan it minted and gated, binds the question's numbers into that plan and computes the answer itself, it has never served a wrong answer on a sealed set: 0 in 118,000 serves (V1.6), 0 in 112,170 (V1.7), 0 binder-attributed wrongs in 11.4 million evaluations (V1.8), 0 in 133,450 across four arms plus 29,941 on the model-free arm (V1.10 SEAL). The single exception in the arc is one plan on V1.10's second sealed sample that computed weight where force was asked, because the plan's cue key did not carry the dimension of the ask.

2. **The reach of that mechanism is bounded by structure, not vocabulary.** On generated traffic it closes the universe (V1.1, 100%). On real word problems it serves 35 to 38 percent of a mixed workload (V1.10), almost entirely two-to-three-step narratives and single-formula physics. The gap to the rest is compositional: knowing a story contains 47, 41 and 20 is not knowing the answer is 47 minus 41 plus 20 (V1.5).
3. **A typed boundary between reading and computing is where safety lives.** The parser proposes structure, a gate checks it against dimensional and role constraints, and only the engine computes. Removing the integrity layer raised adversarial wrong-fire from 0 to 6.6 percent (V0.11, V1.0); removing the gauntlet cost coverage rather than correctness, which is the defense-in-depth result.
4. **Role-aware reuse fixes positional reuse.** V1.7 traced every one of its wrong answers to plans replayed by position rather than by role; V1.8′s role-aware composer produced 0 binder wrongs in 11.4 million evaluations while retaining 79 to 96 percent of coverage.
5. **Scale is monotone in the right direction.** More capable models mint fewer wrong plans (V1.6: 16, 3, 1, 1 wrong-kills from Haiku to Fable) and, in V1.10, have fewer genuine certified wrongs (Haiku 6, Sonnet 3, Fable 1, Opus 0) and more reach. The scale inversion seen in V1.2 was an artefact of mining without gates.
6. **Errata are a measurable floor.** On GSM8K-train and SVAMP, 1 to 2 percent of by-gold wrongs are the corpus, not the engine, and the classes recur across arms and versions (the same dozen items appear in V1.9 and V1.10). Publishing the raw count and the classification side by side is what makes the genuine count readable.
7. **Energy and tokens can be metered honestly, and the answer is not what the premise assumed.** With batch metering and positive controls, a certified serve costs 21 mJ (V1.2) to 0.110 J (V1.8) against about 1,900 J for a local 8B model call; but a pipeline that asks the model for structure per item costs 20 to 38 times a plain call in energy (V1.4) and 2 to 3 times in tokens (V1.10).

What is falsified

Each item below was a pre-registered hypothesis that a sealed run rejected. They are the arc′s main contribution, because each one is a plausible design that people keep proposing.

Idea	Where tested	What happened
Two samples of one model corroborate each other	V1.3, V1.4	Samples share a reading prior; on 240 sealed real items all four arms certified zero, and two samples gave structurally different states three times in four
A verifier can catch the model's computational slips	V1.6	Impossibility checks reached 3.0 to 6.7 percent recall at zero false flags on four scales; catching slips needs recomputation, which is the exact path already
The model's own honesty check filters its wrong readings	V1.9	Honesty gates removed 0 by-gold wrongs on the large arms while killing 13 to 17 percent of correct admissions
Injecting certified values into the model's context improves its answers	V1.3	57.3 percent wrong versus 36.8 percent for plain retrieval; the guarantee belongs to whoever computes
Plan reflexes: reuse a plan because it matches the text	V1.3	7 served, 7 wrong; checks verify a plan matches the text, never that it answers the question
Cue-keyed library reuse without the model	V1.9, V1.10	Would have served 35 to 76 with 13 to 58 wrong (V1.9); at 400k the library's plans on a key disagree on 88 to 93 percent of items and its lone value is right 2 to 27 percent of the time
Agreement between library and the model's own answer is a safety filter	V1.10	The served value is the model's answer by construction; 3 genuine certified wrongs on Sonnet, 6 on Haiku; A2S wrong rate tracks own-answer accuracy (Opus 0 percent at 98.7 percent own-correct, Haiku 3.2 percent at 72.8)
Key consensus of three independent plans is pure	V1.10	Shadow-served 1 on Haiku, wrong; 3 on Fable REPLAY, 1 wrong; 0 served on Sonnet and Opus

Idea	Where tested	What happened
A cheap worker plus selective escalation saves energy	V1.5	5 certified, 2 wrong (40 percent) against a 2 percent bar; coverage 0.63 percent, so saving equals coverage
A relational representation is the hard part	V1.4	The representation detected semantic errors 100 percent at 0 false positives; producing it from a model certified nothing
Dimensional gates certify story semantics	V1.2, V1.5	A dimensionally valid plan can be semantically wrong; a namer assigned a payback-months dimension 131 times to unrelated quantities and passed the gate
Learning curves are flat	V0.11, V1.10	V0.11′s 25k-traffic curve was exactly flat; V1.10′s acquisition curve is concave and non-flat, but only because later tiers supply the keys
The engine in front saves tokens on single-pass traffic	V1.10	The model still answers 62 to 65 percent of queries; counting reading proposals the pipeline costs 197 to 324 percent of a plain model call

V1.10 in detail

V1.10 tested the strongest idea left after V1.9: keep the library, but let it only suggest a reading, and serve that reading only when the engine′s execution of it agrees with the model′s own answer (rule A2S). The pre-registration required zero genuine certified wrongs on the Sonnet arm, at most 1 percent wrong by gold per arm, 30 percent reach on the real-corpus tiers, and tokens per query at half a plain model call.

Workload. 387,062 items in eleven tiers: one-step and multi-step narratives (T1, T2), GSM8K-train (T3), SVAMP (T3′), model-authored 4-to-6-step stories with distractors (T4), single-formula physics (P1), physics chains of 3 and of 5-to-6 nodes (P2, P3), model-authored physics stories (P4), SciQ concept questions that must never be certified (PC), and distractor items with no answer (D). Split by hash into acquisition, development, SEAL and REPLAY. The model answered a stratified sample of 3,042 SEAL items per arm; the engine ran on all 92,392.

Results on SEAL, once per arm (served / wrong by gold / genuine after classification):

Arm	Exact path (model-free)	A2S suggestion	Fresh reading (A2)	Own answers correct	KC shadow
Sonnet	33,102 / 0	410 / 10 / 3	1,417 / 17 / 2	97.9%	0 served
Opus	33,192 / 0	335 / 0 / 0	1,404 / 8 / 0	98.7%	0 served
Haiku	32,135 / 0	283 / 9 / 6	530 / 6 / 2	72.8%	1 / 1 wrong
Fable	35,021 / 0	316 / 2 / 1	1,403 / 8 / 0	98.4%	7 / 0 wrong

REPLAY reproduced every class: A2S wrongs of 3, 3, 2 and 3 per arm, all corpus errata except one Haiku genuine, and one model-free exact wrong on a P1 item on every arm (a plan computing weight for force under a shared cue key).

Reach. Measured on the answered sample as the share of items the exact path missed that the suggestion served: 24.2, 17.6, 12.0 and 17.2 percent over GSM8K and SVAMP for Sonnet, Opus, Haiku and Fable. By tier the suggestion is a two-to-three-step-narrative mechanism (47 to 63 percent on T2) and does nothing on authored stories (0 to 5 percent on T4, 0 on P4).

Tokens. Marginal tokens per item, after subtracting each agent's fixed overhead, are 189 to 247 for an answer and 392 to 1,001 for a reading proposal. The exact path serves 35 to 38 percent of traffic, so the model still answers 62 to 65 percent; counting the readings, the pipeline costs 197 to 324 percent of a plain model call.

Outcome. E on the frozen ladder: the genuine bar fails (3 on Sonnet) and the model-free chain bar fails (the REPLAY P1 item). Had those held, reach would have given D and the by-gold and token bars would have given C. Of the twelve pre-registered hypotheses, five held (physics stories only wrong through reading, concept items never certified, key consensus impure, scale monotone, acquisition curve non-flat) and seven failed.

Can a universal answer filter exist

No, not from any signal this system has, and the reason is structural rather than a matter of tuning. A filter can only reject what it can independently recompute or bound; everything else it can only vote on, and every available voter shares the model's prior.

A post-hoc probe on V1.10's burned SEAL sample (exploratory, not a sealed result) made this concrete. For each answered item on the real-corpus tiers, the library's plans on the item's cue key were executed without consulting the model:

Arm	Model wrong	Plans on the key disagree	Single library value	Single value right	Disagreement catches model wrongs	Disagreement flags right answers
Sonnet	43 of 1,692	1,201 of 1,289	88	2	2 of 43	84
Opus	20	1,146 of 1,259	113	20	2 of 20	91
Haiku	343	1,108 of 1,209	101	21	14 of 343	67
Fable	22	1,135 of 1,258	123	33	3 of 22	87

The library does not know the answer: its plans on a key produce different numbers on about nine items in ten, and its lone value is right 2 to 27 percent of the time. Agreement between library and model therefore selects the one plan that reproduces the model's number, which is why the suggestion rule's accuracy equals the model's. Disagreement catches 4 to 10 percent of the model's wrong answers while flagging about 5 percent of its right ones.

What survives is small and sound: recomputation from a verified structure, and constraint rejections such as a non-integer answer to a count question, which fired 16 times across the four arms and was never wrong. Both have low reach and no false rejects. A verdict interface with four states (certified, rejected, consistent, unknown) is buildable and honest, but its consistent state carries the model's accuracy and no more.

Open problems and the next experiment

Four bottlenecks bind, in order of how hard they are.

1. **The filter is the model.** Any serve-time decision that consults the model inherits its error rate. No engine rule changes this; only recomputation from a structure the engine verified escapes it.
2. **Model-free reach is stuck at the V1.8 level**, about 36 percent of a mixed workload, all of it short narratives and one-or-two-step physics. Nothing in V1.9 or V1.10 moved it.
3. **Structure proposals cost more than answers.** A reading costs 2 to 4 times an answer and single-pass traffic never amortises it. Reuse at zero wrong is what would amortise it, and bottleneck 1 forbids reuse through the model.
4. **Cue keys do not carry dimensions.** A stored plan is not checked against the ask's dimension before it is served. This one is an engineering fix and is first on the list.

Two secondary limits: the physics chain library is shallow (5-to-6-node chains certify a quarter of the time because plans come from shorter donors), and gold quality in public corpora puts a 1 to 2 percent floor under any by-gold measurement, so the genuine count is the only discriminating number.

If model-level error is acceptable, the design changes and the sealed data already say what it yields: serve the exact path with no model call, escalate everything else, and drop reading proposals from the serve path. That answers about 36 percent of traffic at zero model tokens and zero wrong, and the rest costs what a model call costs today. A library serve without the model is not on that list: at 400k scale its lone value is wrong three times in four.

The next experiment (V1.11, not yet specified or frozen) would keep the zero-wrong bar for anything called certified, fix the dimension check, and attack reach by growing what the engine can recompute without the model at serve time: deeper physics chains from verified sub-plans, and story compositions minted only from plans that the engine executed correctly on a held-out set before being admitted. It needs a fresh sealed sample, since SEAL and REPLAY are burned.

Reproducibility and claims

Every version is a git tag on the reflex-engine repository (v1.0-final through v1.10-final, with the pre-registration tags such as v1.10-plan beside them), a frozen directory snapshot that excludes credentials, a final report in Markdown and PDF, and a numbered bug ledger. Sealed datasets are hashed in a manifest before any run; V1.10's hashes cover the source corpora, the sampled items, the post-authoring and post-QC files, and the SEAL and REPLAY sample ids. Each version reproduces from a single command on its tagged commit.

Claimed: the zero-wrong property of recomputation from verified structure at the scales stated; the reach numbers as measured on the stated samples; the negative results in the table above, each with its sealed run; scale monotonicity across four Claude models; the token and energy figures as metered, with their controls.

Not claimed: any property of items the model never saw (the model answered 3.3 percent of V1.10's SEAL; the exact path ran on all of it); any energy or latency number for V1.10 (none was measured that version); generalisation beyond quantitative word problems, physics chains and concept questions in English; any statement about models other than the four Claude arms and the one local 8B model used in early versions.

Models were reached only through sandboxed agents that received batch files of questions. No gold answer, verifier or engine code was ever visible to a model, and usage was taken from the harness, never from the model's own report.

Glossary

Term	Meaning
Reflex	A stored, reusable piece of structure (a parse, a plan over slots, or a physics chain with roles) minted from traffic and gated before reuse
Exact path	The model-free route: match a question to a verified plan, bind its numbers, compute. The only zero-wrong mechanism in the arc
Certified	Served by the engine's own computation with a certificate; the object of every safety bar
Escalated	Passed to the model; the model's answer is returned as the model's, with no claim
Abstain / UNKNOWN	The engine declines; always preferred to a wrong certified answer
Reading	A model's proposal of how a question's answer is computed from its stated numbers (slots language for narratives, chain language for physics)
A2	A fresh reading admitted because its execution agrees with the model's own answer and passes the gates
A2S	A library-suggested reading served on the same agreement rule (V1.10's tested rule); A0S is the same without the honesty gates
KC	Key consensus: three independent library plans agreeing; run only in shadow
Cue key	The structural key on which library plans are indexed and suggested
Wrong by gold	Disagreement with the corpus answer, before any classification
Genuine / corpus / interpretive	Post-hoc classes of a by-gold wrong: the served value is wrong by the text; the gold is an erratum or a corpus convention; both readings are defensible. Golds are never relabelled
SEAL / REPLAY	The two sealed samples of V1.10, hashed before any run and scored once per arm

Term	Meaning
E-SAMPLE	The stratified subset of a sealed split that the model answered; reach is measured on it
Outcome ladder	The pre-registered mapping from bar results to a letter, F (worst) to A (best), frozen before the run
Bug ledger	Numbered defects (B) and notes (N) with preserved superseded runs; nothing deleted or relabelled
Bridge	Sandboxed agents that carry batch files to a model and answers back; no gold, engine or code inside